CPSC 304 Project Cover Page

Milestone #: ___3____

Date: _03/09/2025__

Group Number: ___92___

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

CPSC 304: Project Description
Milestone 3: Project Check-In

General

Your meeting will be sometime in the 10 days following the deadline to turn in your deliverables; see the Canvas assignment for how to sign up for a meeting. **In order to give your TAs time to look over your materials, we require your materials to be turned in by the Friday before your meeting. Specifically, the deliverables are due by 9AM Pacific time on the Friday before your meeting.**

During the meeting, your project TA will be looking for the following things: 1. Team members have arrived on time. Given the fairly short duration of the meeting, we will consider anyone who does not arrive by the start of the meeting as late. 2. Equal contribution to the discussion by all group members.
3. Willingness to participate in the discussion.

Feel free to discuss any questions/concerns you have with your TA. If you are uncomfortable with discussing your concerns publicly, you can also email your TA (contact information can be found on Canvas). If your TA does not reply within 2-3 business days, contact your instructor. If the check in meeting starts late, no extra time will be given. In cases where the meeting starts late because of an issue on the part of the teaching team, the meeting will run for the scheduled length of time.

You may wish to update your deliverables for this milestone after meeting with your TA. We ask that you do not do so until the milestone grades have come out. This will help ensure that your TA looks at the correct commit when grading and it will speed up the process of grading.

Graphical User Interface (GUI)

Your project will require a front end (i.e., a GUI) for the user to interact with the database. When designing the GUI, think of it from the perspective of a user who is unfamiliar with Computer Science. The GUI should be user friendly and relevant to whatever topic your group has chosen. For example, "Create New Account" could be the name of a button that runs an “Insert” query and would be more appropriate than “Insert Query Button”. We expect you to produce a simple but adequate interface; it doesn’t have to be fancy. **You should not be creating a text-based user interface. The user should not be typing in a SQL statement at any point in the process**.

You **cannot** use a GUI generating tool.

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Deliverables

Your deliverables should be committed to the CPSC 304 provided repository by the milestone deadline listed on Canvas in order to allow your TA to look at your milestone before you meet with them. You must submit:

1. Your cover page, as usual.

2. A brief (~2-3 sentences) summary of your project. Many of your TAs are managing multiple projects so this will help them remember details about your project. You can reuse the summary from milestone 2.

Our project creates an Animal Adoption Center/Software focusing on keeping track of pets and their care and management. This is essentially the creation of a centralized animal adoption centre database and medical/care history to ensure their proper well-being.

3. **Timeline and task breakdown/assignment**: The breakdown should be at a level of detail that demonstrates that the group has spent time meaningfully considering what there is left to do. Note that we are not asking you to predict every single possible task that you will need to do. We want to see that the group understands the scope of what is left to do and is prepared to accomplish the remaining tasks in a reasonable manner. We warmly recommend reading the milestone descriptions and associated rubrics on Canvas now so you have an idea of what we expect and what your TAs will be looking for during grading.

Each task should be specifically assigned to a particular group member. Unless otherwise stated, it is assumed that all group members will work equally on the project. This does **not** mean that everyone needs to work on each task together. This means that the overall division of the work is equal. If this is not the case, state the work percentage breakdown for each member. This will serve as a written acknowledgement between all group members that there will be an uneven distribution of work. The member who does not do their fair share of work will have a penalty applied to their final project grade.
While each member is not expected to know about every single line of code in the project, it is expected that all members can talk about the overall architecture of the project.
The timeline should contain enough detail for your project mentor to determine that you understand that you need to produce a GUI for your full project. We strongly recommend reading through the description documents for milestones 4 and 5 along with the associated rubrics on Canvas so that you have a clear understanding of what is expected from you for the term project.

Our timeline and task distribution:

Week 1: Database, SQL Script, and Beginning Dynamic Query Implementations:

Jaskarn: Finalize the SQL script:

- Include DROP TABLE, CREATE TABLE, INSERT statements, and constraints.
- Populate each table with test data.

Sophie:

- Begin implementing dynamic queries for: INSERT and UPDATE
- Set up the initial version of GUI elements for interacting with user input.

Krish:

- Start implementing dynamic queries for: DELETE and SELECTION
- Plan how to display results and error messages in the GUI.

Group Collaboration:

- Review the initial SQL script and GUI prototype as a team.
- Agree on interfaces (dropdowns, input fields) for queries that require user inputs.

Week 2: Further Query Development and GUI Implementation:

Jaskarn:

- Implement dynamic queries for: PROJECTION and JOIN
- Ensure the GUI correctly passes user inputs to these queries.

Sophie:

- Develop hardcoded queries for: Aggregation with GROUP BY and HAVING
- Build and test corresponding GUI controls (e.g., buttons, dropdowns).

Krish:

- Create the more complex queries: Nested Aggregation with GROUP BY and DIVISION
- Integrate these queries into the GUI and their execution.

All Members: GUI Enhancements & Error Handling

- Implement consistent error handling throughout the application.
- Test input sanitization and success/failure messages.
- Test each of the various queries through the GUI.

Weeks 3 & 4: Final and Complete Implementation, Testing , and Documentation

All Members: Full Implementation & Testing

- Put together all components (SQL script, queries, GUI, error handling) into one application.
- Perform thorough testing on chosen database environment (ubcCS server).
- Debug and conduct final code reviews to ensure correct query outputs.

All Members: Documentation & Repository Updates

- Add in the cover page.
- Put together the PDF which will include: A short project description, explanation of any schema changes, and a detailed list of SQL queries with file references (including line numbers)
- Compile and update the README file with necessary project information.

Final Week

All Members: Demo Preparation

- Rehearse the Milestone 5 demo so that every team member understands the overall architecture and is caught up with any late changes.
- Confirm all repository deliverables are final and ready for presentation.

Finals Steps Post Demo:

All Members:

- Complete the Milestone 6 peer and self-evaluation forms

4. The deliverables from milestones 1 and 2 have been added to the repository.

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5. Each group member has made a commit to the repository. **Use your UBC provided account to make the commits**. Please ensure that you have configured your repository settings such that your name shows up when you make a commit.

   The commits do not have to be code related. For example, one group member can commit the milestone 1 deliverables, another the milestone 2 deliverables, and the third member the milestone 3 timeline.

   During the meeting, your project mentor may ask each group member to demonstrate that they are able to make a commit. It is in your best interest to get everything set up now. In the event of a group dispute or disagreement over the level of contribution

made by each member, we will be looking at the commit history. Git is an essential skill and this is a great way to get some practice at using it in a group context. We expect to see incremental and consistent progress on the project throughout the rest of the term. In the event that a request for an accommodation or extension is submitted, your commit history will be one of the factors used in the decision-making process.

6. For groups that are using something other than the JavaScript/CS department provided Oracle or PHP/CS department provided Oracle, **every** group member must send the course coordinator an email. Please be sure to cc your project mentor.

   The email addresses for the course coordinator and your project mentor can be found on Canvas.

   The email must contain the following pieces of information:

   a. Your name, student number, and group number.

   b. The exact combination of programming language and relational database

   c. Something stating that you understand this is completely at your own risk and that we will not be able to provide any technical support whatsoever

   d. Something stating that you understand that the teaching team reserves the right to ask you to come in during the exam period to go through your project again.

7. During the meeting, your project mentor will ask **each** team member to demo a working copy of the sample project from their computer **and** demonstrate that you are able to make modifications and view the result of the modifications. This will involve you running the project, executing a few chosen commands, and demonstrating that you know how to check that the change has been reflected in the database. For example,

your TA may ask you to launch the sample project from your computer, make an edit to the code, and then verify that this change has gone through to your sample project. Your TA may also ask you to insert a tuple, and use SQL Plus (or the IDE if your IDE has functionality to show the status of the database) to verify whether the action executed correctly. You should have the sample project and SQL Plus (if you are using it) ready to go before the start of the meeting.

In the event that you are using a tech stack not supported by the course, you need to have enough of your project in place so that you can demonstrate inserting a tuple, deleting a tuple, and updating the value of a tuple. Make sure your code has been

committed to the repository by the deadline. At this point in time, you don't have to use your project schema for the demo. It is sufficient to use a single table during your demonstration as long as you have some way to show that the changes truly are reflected in the database.

8. In the milestone 3 assignment on Canvas, submit the URL to your group's repository.

https://github.students.cs.ubc.ca/CPSC304-2024W-T2/project_a9b4j_w0a4z_y4b7k

9. You do not need to explicitly declare the use of an AI tool for this milestone.

See the milestone 3 assignment on Canvas for the rubric. Refer to the syllabus for information on late submission/penalty rules.

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FAQ

1. **Do all team members have to be present for the meeting?**
   Yes. See the syllabus for more details on the expectations for milestone 3.

2. **I have a group member(s) not doing their part (e.g., slacking, not replying to communications, etc.). What can I do about this?**
   Email your project TA if you have these concerns and the TA can help you resolve it. **The earlier you notify the course staff, the more likely it is we can provide different options to resolve the situation.** There is also an individual peer evaluation at the end of this project where you can express concerns about your project group members.

   If your project TA does not respond within two business days, contact your instructor.

3. **We have changed our minds about the tech stack we want to use. Are we tied to using what we specified in Milestone 1?**
   You do not have to use the tech stack that you described in Milestone 0. We recommend letting your project TA know about the change during the meeting as they may have some advice for you.

4. **How do we start determining what there is left to do?**
   A project can be broken down into the front end and the back end.

   The frontend has to do with what the user interacts with (e.g., buttons, text boxes, etc.).

In the CPSC 210 term project, you were asked to create a graphical user interface (GUI)– the GUI is considered as the front end.

The backend deals with code that "works in the background". For example, the code that makes a call to the database. Once you have the data, the front end can take over with regards to how to display it.

When determining what there is left to do, try looking at the project from the perspective of the front end and the back end and listing all the things that need to be done. For example, if your GUI has a button that will cause a table to refresh its data, you will need to figure out:

a. How to create a button and a table
b. How to place the button and table in the desired configuration
c. How to place data inside the table

d. How to trigger the data refresh once the button is clicked

5. **How should we divide the tasks?**

The two most common approaches are front/back end or end-to-end.

Teams that divide the tasks by front end/back end will divide the team into two. One group will work solely on creating the front end while the other group works on the back end. As both teams work in parallel, they will start linking up the front and back end to create a functional project.

Coding from end-to-end means that each person does a bit of both the front and back end. Teams that choose this approach typically divide the list of queries and each person is responsible for creating their bit of the GUI and ensuring that it is functional/can properly communicate with the database. This approach can provide a more well-rounded experience with databases but does require good communication amongst the team to ensure that everyone's code can integrate properly.

No matter which method you choose, we recommend designing the architecture of your system prior to implementation. It will make it much easier to build your system and integrate your work.

As a side note, if there is one person who is solely responsible for generating the SQL queries, this would not be considered an equal division of work and their final grade for the project will be adjusted accordingly.

6. **I can't clone my repository via SSH. What should I do?**

   If cloning via SSH does not work, you can try using a personal access token to clone your repository.

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7. **Whenever I make a commit, it doesn't show my name.**
   This link might be useful.

   In short, you need to check what email you are using in your local Git.
   You can then change your email like this:
   ```
   git config --global user.email
   "<YOUR_CWL>@students.cs.ubc.ca"
   ```
   (This is the email that GitHub uses to display that commit message, you can check that your email is <YOUR_CWL>@students.cs.ubc.ca by going to your Github Settings -> Email)
   You can also change your user name with:
   ```
   git config --global user.name "<YOUR_NAME>"
   ```